Missing Topics from Handwritten Notes

Whole topics not covered in the books or papers

Topics from your handwritten notes that have no section of their own in the condensed books or the Paper 1–4 answers (at most a passing mention there), grouped by book and chapter. "(check)" marks a point where the handwriting or figure was doubtful.

Book	Title	Topics
BP	Basic Principles	18
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BP Basic Principles

OTHER TOPICS IN THIS BOOK

MRI

▸ **Principle**

- Protons (hydrogen nuclei) are positively charged and have magnetic spin. MRI uses this property to produce images.
- In the neutral state, protons spin haphazardly.
- The magnet's field aligns the protons in a straight line in the direction of the field.
- A radiofrequency pulse applied perpendicular to the field pushes the aligned protons to a higher energy level.
- When the pulse is removed (relaxation), the protons lose energy. The machine processes this released energy into the image.

▸ **Advantages**

- Much anatomical detail even without contrast.
- Multiple planes without moving the patient.
- Contrast agents carry a considerably smaller risk of allergic reaction than CT contrast.

▸ **Disadvantage**

- The radiofrequency transmitter can cause burns if mishandled.

▸ **Types**

- Conventional (T1- and T2-weighted), diffusion-weighted, dynamic contrast-enhanced, and MR spectroscopy or chemical shift MRI.

Nuclear medicine: renal radionuclide agents

▸ **Tc-99m ethylenedicysteine (EC)**

- Less radiation and less time-consuming than DMSA.
- Reliable differential renal function; can substitute for MAG3.
- Lower sensitivity than DMSA for cortical lesions.

Surgical navigation

▸ **Surgical navigation**

- **Definition:** guiding surgery with computer-based technology; also called computer-aided or image-guided surgery.
- The surgeon uses special instruments tracked by the system. This matters most when the instrument tip cannot be seen, as in minimally invasive surgery.
- **Examples:**
  - iPad-assisted puncture of the pelvicalyceal system, which reduces radiation exposure.
  - Electromagnetic tracking of the puncture.
  - Computer-based tracking of ultrasound-guided kidney biopsy or puncture.
  - Marker-based endoscopic tracking during laparoscopic radical prostatectomy or partial nephrectomy, to reduce positive margins.
  - Stereotactic radiosurgery (CyberKnife), a navigation-based treatment.
- **Advantages:** more surgical precision, better ergonomics, lower risk of surgical errors.
- **Disadvantage:** huge, bulky equipment.

Types of research studies

▸ Classification of epidemiological studies

- Epidemiological studies fall into four groups: descriptive, analytical, experimental and integrative.
- **Analytical:** ecological, case-control, cohort and cross-sectional studies.
- **Experimental:** RCTs, non-randomised controlled trials, and uncontrolled trials.
- **Integrative:** systematic reviews and meta-analyses.

▸ Case-control study

- An observational study of persons with the disease of interest (or another variable) and a suitable control group without it.
- Being observational, no intervention is attempted and no attempt is made to alter the course of the disease.
- Retrospective.
- **Advantages:** good for rare diseases, less time consuming, multiple risk factors studied simultaneously.
- **Disadvantages:** data quality is a problem because it is retrospective; a suitable control group can be hard to find.

▸ Cohort study

- An analytical study done to get more evidence to refute or support an association between a suspected cause and a disease.
- **Other names:** longitudinal, incidence and forward-looking study.

▸ Case-control versus cohort

Feature	Case-control	Cohort
Direction	Effect to cause	Cause to effect
Question tested	Does the suspected cause occur more often in those with the disease than without?	Does the disease occur more often in exposed than unexposed individuals?
Timing	Retrospective	Retrospective or prospective
Subjects	Few only	Large number
Speed	Quick results	Long follow-up
Cost	Relatively inexpensive	Expensive
Rare diseases	Suitable	
Yields	Generally only an estimate of relative risk (odds ratio)	Incidence rate, relative risk (RR) and attributable risk (AR)
Role	Usually the first approach to testing a hypothesis	Reserved for testing a precisely formulated hypothesis

Kaplan–Meier plots

▸ Definition and use

- A non-parametric method for estimating the survival function from lifetime data.
- Often used to measure the fraction of patients still alive at a certain time after treatment.
- Named after Edward L. Kaplan and Paul Meier.

▸ Construction

- A series of declining horizontal steps. With a large enough sample, it approaches the true survival function for that population.
- **Data needed per patient (at least two):** status at last observation, and time to the event (or to censoring).

- **x-axis:** length of each horizontal line is the survival duration for that interval.
- **y-axis:** estimated probability of survival.
- **Worked example:** percent survival against years; patients with a "Gene B signature" have lower survival than those with a "Gene A signature".

▸ **Advantages**

- **Main advantage:** includes censored data, so information about patients lost at any time, for any reason, can still be used.
- Improves professionals' understanding of disease processes.
- Useful in clinical practice for counselling patients with different diseases.

Data types and statistical tests

▸ **Data types**

Type	Group	Features and examples	Tests
Nominal	Qualitative (categorical)	Cannot be arranged in ascending or descending order, for example sex or eye colour	Non-parametric
Ordinal	Qualitative (categorical)	Can be counted or ordered ascending or descending, for example cancer stages (I, II, III, IV) or house numbers	Non-parametric
Interval	Quantitative (numerical)	Measured on a scale with each portion equidistant from the next, for example pain rated 1 to 10	Parametric
Ratio	Quantitative (numerical)	Can be compared as multiples of one another, and multiplied or divided	Parametric

▸ **Parametric and non-parametric tests**

- **Non-parametric test:** used for non-metric independent variables. Examples include chi-square, Mann-Whitney U, Fisher's exact and Wilcoxon tests.
- **Parametric test:** makes assumptions about the population parameter. Examples: z-test for large samples; for small samples, paired t-test, unpaired t-test, one-way ANOVA and two-way ANOVA.

▸ **Student t-test**

- **Definition:** any hypothesis test whose test statistic follows Student's t-distribution under the null hypothesis.
- **Requirements:** random sample, unknown population standard deviation, normally distributed population.
- **One-sample:** compares a population mean with a theoretical value.
- **Unpaired two-sample:** compares means of two independent samples, for example blood pressure of patients versus normal persons.
- **Paired:** compares means of two related groups, for example blood pressure before and after a drug in the same patients (n1 = n2 = n).
- **Formulas** (x is the sample mean, mu the population mean, n the sample size):
  - Significance of a sample mean (whether its difference from the population mean is significant or due to chance): t = (x - mu) / (SD x square root of 1/n).
  - Unpaired: t = (x1 - x2) / (SD x square root of (1/n1 + 1/n2)), where x1 and x2 are sample means and n1 and n2 sample sizes.
  - Paired: t = (x1 - x2) / (SD x square root of (1/n + 1/n)).

▸ **P value**

- **Definition:** the probability of finding the difference between two results by chance, that is, the probability of the null hypothesis being true.
- **Null hypothesis:** assumes no significant difference between specified populations.
- **Example:** tossing a coin, the probability of head or tail is 50%, so P = 0.5.

- **Diagram:** on a probability density curve, P is the shaded tail area beyond the observed data point, in the region of very unlikely observations, with the confidence interval marked on the set of possible results.
- The smaller the P value, the smaller the probability that the event occurred by chance.
- As the confidence interval widens, a previously significant value may become insignificant.

▸ Sample size

- **Definition:** the number of participants or observations in a study, usually written n.
- **Formula:** n = Z squared x p x (1 - p) / c squared, where Z = 1.96 for a 95% CI, c is the confidence interval and p the prevalence in previous studies.
- **Influences:** precision of the estimates and the power of the study to draw conclusions.
- **Power:** the probability of finding a statistically significant result.
- Adequate sample size minimises sampling error; larger samples tend to have a smaller margin of error.
- **Law of diminishing returns:** beyond a certain point, increasing sample size no longer reduces sampling error.

2 Oncology, DSD, Gender & Ageing

CH 24 · BASIC PRINCIPLES OF IMMUNOLOGY AND IMMUNOTHERAPY IN UROLOGIC ONCOLOGY

Immunotherapy

▸ **CIK (cytokine-induced killer) cell therapy**

- **Preparation:** monocytes are isolated from a blood sample and incubated with interferon-gamma, anti-CD3 antibody and recombinant human IL-1 and IL-2. The CIK cells are amplified, quality-tested and infused, where they selectively kill infected or cancer cells.
- Can be autologous or allogeneic.

▸ **Monoclonal antibodies**

Type	Human content	Suffix
Murine	0%	-omab
Chimeric	65%	-ximab
Humanised	More than 90%	-zumab
Fully human	100%	-umab

- **Immunogenicity:** falls from high (murine) to low (fully human).
- **Common side effects:** allergic reactions, flu-like symptoms, nausea.
- **Rare serious reactions:** infusion reactions, leucopenia, skin problems, bleeding.

CH 25 · MOLECULAR GENETICS AND CANCER BIOLOGY

Circulating tumour cells (CTCs)

▸ **Biology**

- Shed from tumours as an early step in blood-borne metastasis; scarce in most tumours (less than 10/mL).
- **Entry into vessels:** EMT-mediated (epithelial–mesenchymal transition) active entry, or non-EMT-mediated passive entry.
- **Metastatic cascade:** intravasation, transit to distant organs, extravasation and invasion, then adaptation and proliferation into a metastasis.
- **Colonisation:** CTCs escape immune surveillance, extravasate and invade the foreign tissue parenchyma to form a metastasis.

▸ **Detection and use**

- Could be used for early tumour detection (CTC-positive versus CTC-negative).
- **Physical methods:** size (filtration), density (gradient centrifugation), electrical properties (dielectrophoresis).
- **Biological methods:** CellSearch system, CTC-chip.

MicroRNA (miRNA)

▸ **Structure and nomenclature**

- Found only in eukaryotic cells; important in gene regulation (gene silencing).
- Single stranded; binds the 3' untranslated region (3' UTR) of its target mRNA.
- **Nomenclature:** "mir" is the precursor; "miR" is the mature form.
- **Gene location:** in introns, or within exons of non-coding genes.

▸ **Biogenesis and action**

- **Biogenesis:** RNA polymerase II transcribes the miRNA gene into pri-miRNA, processed to pre-miRNA in the nucleus, then to mature miRNA in the cytoplasm.
- Mature miRNA joins the RNA-induced silencing complex (RISC) to form the RNA-induced silencing unit, which targets the 3' UTR of mRNA and degrades it.
- Silences genes by mRNA degradation and inhibiting translation.
- Can act as a tumour suppressor gene or as an oncogene.
- Can modulate an entire gene programme, affecting hundreds of transcripts and shutting down entire deregulated pathways.

▸ **Diagnostic and prognostic roles**

- May have a role in disease diagnosis and management.
- May help define cancer subtypes (for example, basal versus luminal breast cancer) and the developmental origin of tumours.
- **Prostate cancer:** miR-143 and miR-187 are potential urine-based markers; miR-187 is an independent predictor of prostate cancer on biopsy.
- **Prognosis:** for example in bronchogenic carcinoma; miR-141 has a role in tumour prognosis.
- **Other diseases:** miR-122 in sepsis; roles also in cardiovascular disease, inflammation, diabetes and perioperative organ protection.

▸ **RNA-based therapy**

- **RNA interference therapeutics:** siRNA and shRNA (target a single transcript), and miRNA.
- **Single-transcript example:** PSMA-based RNA therapy in prostate cancer.
- **miRNA replacement therapy:** drug mimics of the endogenous miRNA, for example miR-16.
- **miRNA inhibitor therapy:** drugs that target the miRNA, for example miR-122.

OTHER TOPICS IN THIS BOOK

Next-generation sequencing (NGS)

- Also called second-generation or massively parallel sequencing.
- A DNA and RNA sequencing technique distinct from first-generation (Sanger) sequencing.
- **DNA sequencing:** determining the precise order of nucleotides in a DNA molecule.
- **Advantages:** low cost, speed, high yield.
- **Clinical uses:** sequencing entire cancer genomes, detecting pathogens, discovering new tumour markers, providing accurate prognostic and diagnostic data.

3 Infections & Inflammation

CH 26 · INFECTIONS OF THE URINARY TRACT

Natural defence against UTI

▸ **Uromodulin (Tamm–Horsfall protein) as a UTI defence**

- Saturates the mannose-binding sites of type 1 pili, blocking bacterial binding to the urothelium.

Chronic pyelonephritis

- Occurs with an underlying structural or functional abnormality; uncomplicated chronic bacterial pyelonephritis is very rare.
- **Symptoms:** none unless renal insufficiency develops, then those of chronic renal failure. May have a history of recurrent acute pyelonephritis.
- **Gross:** diffusely contracted, scarred, pitted kidney with Y-shaped scars; thin parenchyma with loss of corticomedullary differentiation.
- **Histology:** patchy interstitial infiltrate of lymphocytes, plasma cells and polymorphs, with periglomerular fibrosis. Leucocyte and hyaline casts in tubules resemble thyroid colloid (renal thyroidization).
- **Pyelography:** asymmetric, irregular outlines with blunting and dilatation of one or more calyces.
- **CT:** parenchymal thinning or atrophy, calyceal clubbing or asymmetry.
- **Function:** nuclear renogram.
- **Management:** treat infection with prolonged therapy, prevent future infection, preserve renal function. Relieve obstruction and remove stones, which may enhance damage. Arrange urological and nephrological evaluation.

CH 27 · INFLAMMATORY AND PAIN CONDITIONS OF THE LOWER GENITOURINARY TRACT: PROSTATITIS, ORCHITIS, EPIDIDYMITIS AND SCROTAL PAIN

Classification and clinical features

▸ **Drach classification of prostatitis**

- Acute bacterial, chronic bacterial, non-bacterial prostatitis and prostatodynia.
- NIH IIIA corresponds to non-bacterial prostatitis; IIIB to prostatodynia.

CH 29 · SEXUALLY TRANSMITTED DISEASES

Classification of Anti-HIV Drugs

Class	Drugs
Nucleoside reverse transcriptase inhibitors (NRTIs)	Zidovudine, lamivudine, stavudine, didanosine, zalcitabine, abacavir
Nucleotide reverse transcriptase inhibitor	Tenofovir
Non-nucleoside reverse transcriptase inhibitors (NNRTIs)	Nevirapine, delavirdine, efavirenz
Protease inhibitors	Nelfinavir, amprenavir, saquinavir, indinavir, ritonavir, lopinavir, atazanavir, fosamprenavir
Entry (fusion) inhibitor	Enfuvirtide

CH 31 · TUBERCULOSIS AND PARASITIC INFECTIONS OF THE GENITOURINARY TRACT

Drugs in GUTB and Dosage

▸ Streptomycin

- **Action:** binds 16S rRNA of the 30S ribosomal subunit, causing codon misreading and inhibiting protein synthesis (bactericidal).
- **Side effects:** vestibular and auditory ototoxicity, nephrotoxicity, injection-site abscess, thrombocytopenia, aplastic anaemia, haemolytic anaemia, agranulocytosis.

OTHER TOPICS IN THIS BOOK

Antibiotics

▸ Penicillins

- **Action:** bind penicillin-binding proteins and inhibit transpeptidase, blocking cross-linking in cell wall synthesis.
- **Adverse effects:** local irritation; hypersensitivity in 1-10% including anaphylaxis; rarely superinfection.

Group	Drugs
Natural prototype	Penicillin G
Acid-resistant alternative to penicillin G	Phenoxymethylpenicillin (penicillin V)
Penicillinase-resistant	Methicillin, cloxacillin, dicloxacillin
Extended-spectrum: aminopenicillins	Ampicillin, amoxycillin, bacampicillin
Extended-spectrum: carboxypenicillins	Carbenicillin, ticarcillin
Extended-spectrum: ureidopenicillins	Piperacillin, mezlocillin

▸ Piperacillin–tazobactam

- **Piperacillin spectrum:** E. coli, Klebsiella, some Bacteroides, Gram-positive organisms; frequently used for serious Gram-negative infections.
- Concurrent gentamicin or tobramycin is advised with piperacillin.
- Tazobactam is paired because its pharmacokinetics match piperacillin's.
- **Dose:** 4.5 g (4 g/500 mg) IV every 6–8 hours; adjust when creatinine clearance is below 40 mL/min.
- **Side effects:** nausea, constipation or diarrhoea, allergic reactions including anaphylaxis, thrombocytopenia.

▸ Imipenem–cilastatin

- Rapidly hydrolysed by dehydropeptidase-I on the brush border of renal tubular cells.
- Usually given with cilastatin, a reversible dehydropeptidase-1 inhibitor that prevents its renal metabolism.
- **Spectrum:** Gram-positive cocci, Enterobacteriaceae, Pseudomonas, Listeria, and anaerobes such as B. fragilis and C. difficile.
- **Dose:** imipenem–cilastatin 0.5 g IV every 6 hours; add gentamicin for Pseudomonas.
- **Side effects:** nausea, vomiting, diarrhoea, skin rashes, hypersensitivity; seizures at higher doses in predisposed people.

ADDITIONS FOUND ON RE-CHECK

SIRS and the older sepsis definitions

▸ **Older (Sepsis-1/2) definitions**

- **Bacteraemia:** viable bacteria in the blood.
- **SIRS:** two or more of the following, without evidence of infection: temperature below 96.8 F or above 100.4 F; heart rate above 90/min; respiratory rate above 20/min (or PaCO2 below 32 mmHg); white cell count below 4000/mm3 or above 12000/mm3 (or 10% bands).
- **Severe sepsis:** sepsis with sepsis-induced organ dysfunction or tissue hypoperfusion, typically systolic BP below 90 mmHg or MAP below 70 mmHg.

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CH 41 · MANAGEMENT STRATEGIES FOR VESICOURETERAL REFLUX

Associated anomalies

▸ **VACTERL and CHARGE associations**

- **VACTERL:** vertebral, anal, cardiac, tracheo-oesophageal, renal and limb anomalies.
- **CHARGE:** coloboma, heart disease, atresia of the choanae, retarded development, genital hypoplasia, ear anomalies.

CH 50 · RENAL DYSGENESIS AND CYSTIC DISEASE OF THE KIDNEY

Genes in kidney and ureter development

▸ **Renin–angiotensin system (AGTR2)**

- Active in fetal life through AGTR2 (angiotensin II receptor type 2); maintains fetal glomerular filtration, ensures urine production and supports normal kidney and ureter development.
- AT1 and AT2 receptors are both in the developing mesonephros and metanephros. AT2 acts in embryonic renal development; AT1 modulates later vascular development and classical angiotensin II effects (vasoconstriction, sodium reabsorption).
- **ACE inhibitors in pregnancy:** oligohydramnios, fetal hypotension, anuria.

CH 52 · ECTOPIC URETER, URETEROCELE AND URETERAL ANOMALIES

Embryology and aetiology

▸ **Ectopic vas (anomalous common excretory duct)**

- A vas joined to the ureter (anomalous common excretory duct) should be considered in males with recurrent epididymitis and ipsilateral hydronephrosis.

CH 63 · PAEDIATRIC UROLOGIC ONCOLOGY: RENAL AND ADRENAL

Neuroblastoma: pathology

▸ **Ganglioneuroma**

- Diagnosed in older children; uncommon in the adrenal.
- Resect only when symptomatic. Survival is not influenced by extent of resection, so avoid aggressive resection.

Nephrogenic rests

▸ **Definition and background**

- **Nephrogenesis:** ureteric bud meets metanephric blastema at 5 weeks; fetal lobulations by 28 weeks; complete at 36 weeks.
- **Definition:** metanephric blastema persisting beyond 36 weeks of gestation, when nephrogenesis is almost complete.
- Found in 1% of infant autopsies.
- Usually clinically occult; found incidentally with a Wilms tumour.

▸ **Diffuse nephroblastomatosis**

- Multiple rests that enlarge the kidney but preserve its shape.
- Prone to Wilms tumour; bilateral lesions are common.

▸ Perilobar versus intralobar rests

	Perilobar	Intralobar
Share	90%	10%
Number and margin	Multiple, well circumscribed, usually smooth	Single or multiple, irregular margins that may interdigitate with adjacent normal interstitium
Main tissue	Blastema	Stroma and epithelium
Malignant transformation	Less chance	More chance
Associations	Beckwith–Wiedemann, Perlman syndrome, trisomy 18	WAGR, Denys–Drash, sporadic aniridia

▸ Pathology

- **Gross:** tan nodules or white plaques.
- **Histology:** sclerosing, hyperplastic, dormant or neoplastic (transforming into Wilms tumour).

▸ Imaging

- **Ultrasound:** nodular appearance, loss of corticomedullary differentiation, diffusely echogenic kidney.
- **CT:** enhance less than parenchyma; surface may look lobulated.
- **MRI (best):** hypointense on T1, hyperintense on T2.
- **Angiography:** rests do not opacify; contrast may puddle in intervening normal parenchyma.

▸ Risk and treatment

- The more rests, the more likely a Wilms tumour.
- **Highest risk:** diffuse perilobar rests, about 35% (check).
- **Treatment:** controversial. Some give chemotherapy; others hold that close radiological surveillance of an enlarging mass is enough.

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CH 32 · EMBRYOLOGY OF THE HUMAN GENITOURINARY TRACT

Metanephros and nephron development

▸ **Cadherins**

- Proper mesenchymal condensation and tubulogenesis need expression of R-cadherin, E-cadherin and cadherin 6, and suppression of cadherin 11.

▸ **Larsson stages of nephron development**

Stage	Features
I	Mesenchymal condensation forms a pretubular aggregate (a detached oval mass), which becomes the renal vesicle, still unconnected to the ureteric bud.
II	Two clefts develop.
III	Glomerular (Bowman) capsule oval; proximal and distal tubules identifiable; endothelial and mesangial cells recruited to form the glomerulus.
IV	Glomerulus round, resembling a mature renal corpuscle; proximal tubule appears mature; distal segments still primitive.

- In humans, all nephrons at birth are at varying steps of stage IV.
- More differentiated nephrons lie near the juxtamedullary region; less differentiated ones lie at the periphery.
- Renal maturation continues after birth, although nephrogenesis is complete before birth.

CH 43 · EXSTROPHY–EPISPADIAS COMPLEX

Surgical approaches in the newborn

▸ **Warsaw approach**

- **Primary closure:** bladder, posterior urethra, pubis and abdominal wall, with or without osteotomy; spica cast for 3 weeks, then elastic bandage for 3 weeks.
- **Second stage:** bladder neck and epispadias repair together, once bladder capacity is suitable and the child is interested in continence.
- **Complications:** fistula or urethral stricture in 10%.

CH 46 · NEUROMUSCULAR DYSFUNCTION OF THE LOWER URINARY TRACT IN CHILDREN

Management

▸ **Artificial somatic–autonomic reflex pathway**

- A novel approach, with limited data, aiming to restore bladder function in children.
- Made by limited laminectomy and microanastomosis of a lumbar ventral root to the S3 ventral root.

6 Female Urology

CH 114 · CONSERVATIVE MANAGEMENT OF URINARY INCONTINENCE: BEHAVIOURAL AND PELVIC FLOOR THERAPY, URETHRAL AND PELVIC DEVICES

Mechanical Vaginal and Urethral Devices

▸ **Intraurethral devices**

- **Contraindications:** urge incontinence (device not retained), active UTI, urethral stricture; not useful in a fibrotic urethra.

▸ **FemSoft insert**

- **Design:** single-use silicone tube in a mineral-oil-filled silicone sleeve.
- **Use:** remove for voiding; change every 6 hours.
- **Sizes:** 16, 18 and 20 Fr; lengths 3.5 and 4.5 cm.

CH 121 · URINARY TRACT FISTULAE

Ureteroenteric and Pyeloenteric Fistula

▸ **Ureteroenteric fistula**

- **Commonest cause:** Crohn disease of the right ureter and terminal ileum, so most are right-sided.
- **Diagnosis:** retrograde pyelogram (RGP), CT or MR.
- **Treatment:** ureterolysis with stenting and possible bowel resection; ureteric resection may be unnecessary if the ureter separates from the bowel.

OTHER TOPICS IN THIS BOOK

Female Urethral Occlusive Devices

- **Generations:** first, Reliance (no longer used); second, FemSoft insert.
- **Suited to:** mild SUI during sports and women unwilling to have SUI surgery.

7 Voiding Dysfunction

CH 109 · NEUROMUSCULAR DYSFUNCTION OF THE LOWER URINARY TRACT

Spinal cord anatomy

- 31 segments, each with paired ventral and dorsal roots that combine on each side to form a spinal nerve.
- **Vertebral to cord levels:** C1–C7 to C1–C8; T1–T8 to T1–T12; T9–T11 to L1–L5; T12–L1 to S1–S5.
- **Conus medullaris:** tapered distal cord at L1–L2.
- **Filum terminale:** pia mater extending to the coccyx; stabilises the cord.
- **Cauda equina:** L2–S5 and coccygeal roots.

Classification

▸ **Lapides classification**

Type	Cause	Features
Sensory	Diabetes, tabes dorsalis, pernicious anaemia; interrupted sensory fibres between bladder and cord, or afferent tracts to the brain	Sensation impaired early; late decompensation to a large-capacity atonic bladder
Motor paralytic	Extensive pelvic surgery or trauma destroying parasympathetic supply	Early painful retention or inability to initiate or maintain micturition; end stage decompensation
Uninhibited	Lesions in or above the brainstem: stroke, Parkinson disease, brain tumour	
Reflex	Suprasacral spinal cord injury, transverse myelitis	
Autonomous	Lower motor neuron lesion destroying sacral cord, sacral roots or pelvic nerves	Complete motor and sensory separation of bladder from sacral cord

Non-neurogenic and functional voiding disorders

▸ **Bashful syndrome**

- Low-flow, low-pressure voiding in young men with marked hesitancy when others are present.
- Psychological: a tendency to overcontrol micturition.
- **Treatment:** behavioural therapy or drugs.

CH 110 · OVERACTIVE BLADDER

Evaluation

▸ **OAB Symptom Score (OABSS, Homma et al)**

- Four questions on urinary condition during the last week.

Question	Scoring
Q1 Daytime voids (waking to sleeping)	7 or fewer: 0; 8–14: 1; 15 or more: 2
Q2 Night-time voids (sleeping to waking)	0: 0; 1: 1; 2 or more: 2
Q3 Urgency (sudden desire difficult to defer)	Not at all 0; less than once a week 1; once a week or more 2; about once a day 3; 2–4 times a day 4; 5 or more times a day 5
Q4 Urgency incontinence (leak because the desire cannot be deferred)	Same options, 0–5

CH 111 · UNDERACTIVE DETRUSOR

Management

▸ **Intravesical electrotherapy**

- **Method:** saline-filled bladder; current through an electrode at the catheter tip, skin pad as return electrode.
- **Schedule:** daily sessions, usually 1 hour or more; 10–15 sessions as a trial.
- **Mechanism:** activates A-delta afferents, giving central reflex detrusor activation.
- **Drawbacks:** very time-consuming, resource-heavy, no definitive safety or application studies.
- **Results:** promising early results, mainly in children; a sham-controlled trial was negative.

CH 113 · PHARMACOLOGIC MANAGEMENT OF LOWER URINARY TRACT STORAGE AND EMPTYING FAILURE

Drugs for storage failure

▸ **Flavoxate**

- Mixed actions, unknown mechanism: calcium antagonist, phosphodiesterase inhibitor, local anaesthetic.
- Used in detrusor overactivity or OAB; no RCTs.

Drugs to facilitate emptying

▸ **Prostaglandins to aid emptying**

- Only a paucity of literature for increasing bladder contractility.

▸ **Baclofen**

- GABA-B agonist (GABA is inhibitory); spinal inhibitory action controls micturition disorders and lowers outlet resistance.
- Not effective orally for voiding dysfunction, as it does not cross the blood-brain barrier.
- Intrathecal baclofen is an option to lower outlet resistance; few studies.

OTHER TOPICS IN THIS BOOK

Storage failure at the outlet: additional therapies (Ch 87)

▸ **Gracilis myoplasty (urethral neosphincter)**

- Autologous gracilis transfer forming a neosphincter around urethra and bladder neck; large studies lacking.

▸ **Bladder outlet closure**

- **Indications:** failure of other surgery, or urethral or bladder neck erosion or destruction.
- **Functional:** autologous pubovaginal sling obstructs the bladder neck but allows a pop-off leak at higher pressures.
- **Complete:** when a functional outlet cannot be kept; multilayered bladder neck closure (transvaginal, transabdominal or combined) with an abdominal stoma for CIC.

Emptying failure: catheters and manoeuvres

▸ **Credé and Valsalva**

- May succeed in an areflexic bladder with low outlet resistance (after surgery or botulinum toxin); these patients have stress incontinence and can empty when they choose.
- Credé can generate over 50 cmH2O intravesical pressure.

- **Credé methods:** open hand, closed fist, or rolled-up towel.
- **Valsalva:** sit with the abdomen protruding onto the thighs; hugging knees and legs may stop the abdomen bulging.

▸ **Triggering reflex contractions**

- In SCI with intact cord reflexes, rhythmic suprapubic pressure (seven or eight pushes every 3 seconds) summates on wall tension receptors and activates the reflex arc.
- **Other triggers:** pulling pubic, scrotal or thigh skin or hair; digital rectal stimulation; squeezing the clitoris.

Collection devices and absorbent products

- No effective external collection devices for women; men can use a condom catheter, penile sheath or Texas catheter.
- **Absorbents:** pads, shields, drip collectors, guards, briefs, undergarments, underpads.
- **Incontinence-associated dermatitis:** surface skin inflammation with redness, oedema, sometimes bullae with clear exudate; mainly in skin folds; may develop candidiasis or bacterial infection.

Diabetes and urology

▸ **FRIENDS mnemonic (urological complications)**

- **F:** Fournier gangrene and soft-tissue infection.
- **R:** Renal failure.
- **I:** Infections and UTI.
- **E:** Enhanced wound complications and emphysematous pyelonephritis (EPN).
- **N:** Necrosis (papillary).
- **D:** Diabetic cystopathy.
- **S:** Stones (low urine pH) and Sexual dysfunction.

8 Nocturia & Male Incontinence

CH 112 · NOCTURIA

Global polyuria: evaluation and treatment

▸ **Overnight water deprivation test**

- **First-morning urine osmolality above 800 mOsm/kg H2O:** normal; points to primary polydipsia (appropriate renal response and ADH secretion).
- **Below 800 mOsm/kg:** give desmopressin 40 µg intranasal or 0.4 mg orally; recheck osmolality 3–5 hours later.
  - 800 or more: central diabetes insipidus (DI).
  - Below 800: nephrogenic DI.

CH 123 · SURGICAL PROCEDURES FOR SPHINCTERIC INCONTINENCE IN THE MALE: ARTIFICIAL URINARY SPHINCTER AND PERINEAL SLINGS

History of continence devices

- **Evolution:** Kaufman I, II and III, then slings, then AMS 721, 742, 791 and 792, then AMS 800.

Male slings

▸ **Bone-anchored bulbourethral sling (InVance)**

- **Mechanism:** urethral compression.
- **Approach:** perineal incision exposing bulbospongiosus and the medial inferior pubic ramus.
- **Fixation:** sling lies over bulbospongiosus, fixed with 3 screws each side to the inferior pubic ramus.
- **Outcome:** cure or improvement 67–92%; failure 8–33%.

OTHER TOPICS IN THIS BOOK

Male urethral compression devices

▸ **External devices (penile clamps)**

- **Examples:** Hiesler metal clamp (no longer used), Cunningham clamp, C-3, U-Tex.
- Cause pressure necrosis, so release every 3–4 hours.
- Cannot be used with impaired penile sensation.

▸ **Internal compression devices (Berry, Kaufman)**

- **Berry's implantable prosthesis:** no longer used.
- **Kaufman I and II:** compressed the bulbar urethra by cavernous–crural crossover (I) or approximation (II).
- **Kaufman III:** silicone gel prosthesis implanted perineally to compress the bulbar urethra.

9 Andrology & Sexual Medicine

CH 70 · SURGICAL MANAGEMENT OF MALE INFERTILITY

Pre-requisites for IVF

▸ **Female partner**

- **Infection screen:** HBsAg, anti-HCV, HIV I and II, HTLV I and II, and syphilis by rapid plasma reagin (RPR) test.
- **Other tests:** Pap smear, blood group, Rh type and antibody screen.
- **Reproductive hormone assay:** FSH and oestrogen. FSH rises when the ovary cannot produce normal eggs; oestrogen influences FSH secretion and relates to egg quality.
- **Sonohysterogram:** assesses the endometrial cavity for polyps, scarring and fibroids.

▸ **Male partner**

- HIV, HBsAg, anti-HCV and RPR, with semen evaluation.

Semen processing techniques

▸ **Simple wash**

- Single or double wash with culture medium, then centrifugation. Simplest method; simple and effective.
- Fails mainly because it generates reactive oxygen species (ROS).

▸ **Swim-up**

- Sperm wash medium is layered over liquefied semen; the tube is incubated at a 45 degree angle for 60 minutes.
- Viable spermatozoa swim up into the medium and are aspirated; non-viable sperm stay behind.

▸ **Swim-down**

- Semen is layered over bovine serum albumin medium and incubated at 37 degrees C for 1 hour. The most motile sperm move downward.

▸ **Density gradient centrifugation**

- **Principle:** each spermatozoon settles at the gradient level matching its own density. Morphologically normal mature spermatozoa have a density of 1.10 g/mL.
- **Method:** semen is layered over a lower-density and then a higher-density solution and spun for 20 minutes at 1600 rpm.
- **Layers, top to bottom:** seminal plasma; white cells and debris; abnormal and non-motile sperm; the interface; viable motile spermatozoa in the pellet.
- **Drawback:** in severe oligozoospermia, sperm are lost in the low-density fraction or at the interface.

▸ **Filtration**

- Glass wool, glass beads or cross-linked dextran gel; gravity helps pull motile spermatozoa through the column. Not used routinely.

Sperm cryopreservation (sperm banking)

▸ **Definitions**

- **Cryopreservation:** the procedure that makes it possible to stabilise and store sperm at cryogenic temperature.
- **Sperm bank:** a place where semen is cryopreserved for later use in artificial insemination.

▸ **Indications**

- Autoimmune disorders or diabetes.
- Donor insemination programmes.
- To overcome the stress of producing a semen sample on the day of treatment.
- Before vasectomy, or at the time of the procedure.
- Exposure to potentially toxic agents that interfere with gametogenesis.

▸ **Cryoprotectants**

- Protect spermatozoa from freeze damage by ice crystallisation.
- Glycerol (most commonly used), ethylene glycol, dimethyl sulphoxide and 1,2-propanediol.

▸ **Freezing techniques**

- **Slow freezing:** manual or automated; 2 to 4 hours in total. Semen is cooled progressively while cryoprotectant is added.
  - Room temperature to 5 degrees C at 0.5 to 1 degree C per minute.
  - 5 to minus 80 degrees C at 1 to 10 degrees C per minute.
  - Store in liquid nitrogen at minus 196 degrees C.
- **Fast freezing:**
  - Mix the sample with an equal volume of cold cryoprotectant.
  - Load into straws and hold at 4 degrees C for 10 minutes.
  - Hold the straws 15 cm above liquid nitrogen (at minus 80 degrees C) for 15 minutes.
  - Immerse in liquid nitrogen at minus 196 degrees C.

▸ **Cryo-injury to sperm**

- **Mechanisms:** thermal shock; ice-crystal formation (impairs organelle function and survival); osmotic shock; cellular dehydration (toxic damage from high solute content).
- **Membranes:** plasma and mitochondrial membranes are injured, activating caspases 3, 8 and 9. This lowers ATP generation and so reduces motility.
- **DNA damage:** from oxidative stress, thermal denaturation and defects in DNA repair enzymes.

▸ **Thawing**

- Done gradually so cells recover normal biological activity without abrupt temperature change. Routine temperature: 37 degrees C.
- **Technique:** 10 minutes at room temperature, 10 minutes in a 37 degree C thermostat, 15 minutes at room temperature, then washing and centrifugation before use.

▸ **Outcome**

- Storage for up to 28 years.
- Long-term follow-up: no increase in birth defects or chromosomal anomalies in offspring compared with the general population.
- Better results with ICSI than with IVF or IUI, because ICSI needs only a small number of motile spermatozoa.
- Pregnancy rates vary with the source (testicular versus epididymal sperm) and the ART method used.

Assisted reproductive technology (ART)

▸ **Third-party assisted ART**

- **Donor sperm:** when the husband has a genetic disease.
- **Egg donation:** when the woman cannot produce eggs. Ova from another woman after ovarian stimulation are fertilised by IVF and the embryo placed in the wife's uterus.
  - **Indications:** ovarian insufficiency, after chemotherapy or radiotherapy, after oophorectomy, women born without ovaries, carriers of known genetic diseases, poor egg quality.
- **Surrogates and gestational carriers:** when the wife cannot carry a pregnancy to term.

- **Surrogate:** inseminated with the male partner's sperm; the child is biologically related to the surrogate and the male partner.
- **Gestational carrier:** receives an IVF embryo from the husband's sperm and the wife's ova; the child is not related to the carrier.

CH 72 · EVALUATION AND MANAGEMENT OF ERECTILE DYSFUNCTION

Oral therapy

▸ **Oral phentolamine**

- Nonspecific alpha-1 and alpha-2 antagonist; relaxes corporal smooth muscle by postsynaptic alpha-1 blockade.
- About 40% response in mild ED; needs more study.

▸ **Yohimbine**

- Central alpha-2 antagonist. Best limited to psychogenic ED; does not enable intercourse in organic ED.

▸ **Apomorphine**

- Role in psychogenic ED. Sublingual 2, 4 or 6 mg; no erectile efficacy if swallowed.

▸ **Trazodone and melanocortin agonists**

- **Trazodone:** its active metabolite activates the pro-erectile 5-HT2C receptor and antagonises the anti-erectile 5-HT1A receptor.
- **Melanocortin receptor agonists:** early studies show efficacy in inducing erection.

▸ **Herbal and dietary supplements for erectile dysfunction**

- Herbs, dietary supplements, ginkgo biloba, L-arginine (substrate for NO synthesis), Korean red ginseng.

Topical therapy

▸ **Topical agents for erectile dysfunction**

- **Alprostadil 0.3%:** significantly more successful intercourse than placebo.
- **Prostaglandin E1 ethyl ester:** a prodrug option.
- **Nitroglycerine 2%:** headache; only rarely a rigid penis.
- **Papaverine:** minimal absorption because of large molecular size.
- Topical and transdermal agents still need further trials.

CH 76 · DIAGNOSIS AND MANAGEMENT OF PEYRONIE DISEASE

Non-surgical treatment

▸ **Electromotive drug administration (EMDA)**

- Bypasses hepatic metabolism and raises drug concentration in target tissue.
- Verapamil alone: no benefit. Verapamil plus dexamethasone: decreased plaque volume and curvature.
- **Adverse effect:** temporary erythema at the electrode site.

▸ **Radiotherapy for Peyronie disease**

- Anti-inflammatory effect; no clinical benefit. Risks: malignancy, and ED in the elderly.

▸ **Topical drugs and combination therapy**

- **Topical drugs tried:** beta-aminopropionitrile, superoxide dismutase, verapamil, trifluoperazine. None effective.

- **"Three-armed approach":** daily pentoxifylline plus L-arginine, verapamil injections twice weekly, daily traction. Best opportunity for deformity correction.

OTHER TOPICS IN THIS BOOK

Unconsummated marriage

- **Definition:** to consummate a marriage is to complete the married relationship by sexual intercourse.
- **Frequency:** common; many sexologists report about every third patient has this problem.
- **Effects:** severe stress and frustration, hostility, even divorce, ruined lives.
- Treat as a semi-emergency, because delay builds frustration and hostility that hinder treatment and understanding.
- Cases with deep-rooted psychopathology are difficult and most often fail.

▸ Aetiology

- **Ignorance:** many young couples, though educated, are ignorant about sexual practice; false expectations or wrong technique lead to failure.
- **Fear of pain:** a preconditioned notion in many women that sex is very painful leads to severe vaginismus, making intercourse impossible.
- **Performance anxiety with erectile failure:** may stem from myths such as masturbation guilt and Dhat syndrome.
  - **Dhat syndrome:** culture-bound; the man complains of premature ejaculation or impotence and believes he passes semen in his urine, with no organic cause.
- **Personality disorders:** phobias, fear and depression affect the couple's sexual life.
- **Physical defects:** women, local inflammatory pathology and rarely a tough hymen; men, severe penile curvature or veno-occlusive disease causing inadequate erections.
- **Social factors:** family pressure, lack of privacy and job stress can make intimacy impossible.

▸ Treatment

- **Basic sex counselling:** correct misconceptions, allay fears, teach the right technique.
- **Social counselling:** correct excessive in-law interference, create privacy, minimise stress, improve communication between wife and husband.
- **Vaginal self-dilatation:** when fear of pain is the main problem.
- **Surgical hymenectomy:** for a tough hymen.
- **Temporary medical support:** sildenafil or intrapenile injections for performance anxiety; after a few successful intercourses men gain confidence and usually no longer need it.
- **Correction of physical defects:** severe chordee, circumcision for phimosis, penile revascularisation, penile prosthesis.

10 Testis, Penis & Urethra

CH 79 · NEOPLASMS OF THE TESTIS

Extragonadal GCT

▸ **Site and behaviour**

- Arises more often in the mediastinum than the retroperitoneum; seminoma is the commonest type.
- **Retroperitoneal GCT:** behaves like testicular GCT in biology and prognosis.
- **Mediastinal seminoma:** same prognosis as testicular seminoma.
- **Mediastinal mature teratoma:** low malignant potential; generally cured surgically.

▸ **Mediastinal NSGCT**

- Less chemosensitive than testicular or retroperitoneal NSGCT; 5-year overall survival 45%.
- More likely to contain a yolk sac component, so AFP is raised.
- Associated with haematological malignancies carrying extra 12p.

▸ **Diagnosis when suspected**

- Extragonadal mass plus raised AFP and/or hCG plus normal testis examination: enough to start chemotherapy.
- Normal markers and normal testis examination: biopsy the mass to confirm.

Choriocarcinoma syndrome

▸ **Choriocarcinoma syndrome**

- **Features:** very high beta-hCG (above 50,000 IU/L), mediastinal, abdominal or testicular mass, gynaecomastia, hyperthyroidism; can cause tumour haemorrhage.
- Medical emergency needing immediate chemotherapy.
- **Sequence:** first course EP for 3 days with beta-blockers, then BEP or VIP; radical orchidectomy between cycles.

CH 82 · TUMOURS OF THE PENIS

Non-Squamous Malignancies

▸ **Adenosquamous carcinoma of the penis**

- Large firm mass; CEA positive; probably best treated by local excision and surveillance.

11 Stones, Obstruction & Trauma

CH 93 · MANAGEMENT OF UPPER URINARY TRACT OBSTRUCTION

Stone types

▸ **Matrix stones**

- Matrix is the non-crystalline component of stones; predominantly matrix stones are rare.
- Pure matrix stones may be up to 65% protein.
- **Matrix proteins:** osteopontin and calprotectin.
- Beta-2-microglobulin (microfibrillar protein) matrix stones occur in chronic renal failure patients on dialysis.
- **Imaging:** typically radiolucent; may be mistaken for a tumour or a uric acid stone.
- **Treatment:** PCNL is the best choice, but ureteroscopy can also be used.

▸ **Ammonium acid urate stones**

- **Prime risk factor:** conditions causing metabolic acidosis.
- **Further associations:** inflammatory bowel disease, morbid obesity and recurrent uric acid stones.
- **Treatment by cause:**
  - Bowel disease: standard fluid intake, oral calcium, alkalinisation and oxalate reduction.
  - History of uric acid calculi: more fluid, protein and salt restriction, potassium citrate alkalinisation and allopurinol.
  - Laxative abuse: a healthier bowel regimen.
  - Chronic infection: as for struvite calculi.

CH 95 · URINARY LITHIASIS: ETIOLOGY, EPIDEMIOLOGY AND PATHOGENESIS

Hypercalciuria

▸ **Fast and calcium load test**

- Restricted diet for at least 7 days beforehand.
- Drink 300 ml of water at 12 hours and again at 9 hours before the calcium load.
- Two hours before the load, empty the bladder and drink 600 ml of water.
- Collect urine for 2 hours: the fasting urine.
- Give a 1 g oral calcium load as 250 ml of a liquid synthetic diet (Calcitest).
- Collect urine for the next 4 hours: the post-load urine.
- Measure calcium and creatinine in both samples.
- **Normal post-load urinary calcium:** less than 0.2 mg per mg creatinine.

CH 97 · SURGICAL MANAGEMENT OF URETERAL CALCULI

Ureteric calculi

▸ **Stone in a megaureter**

- **Non-obstructed:** medical expulsive therapy (MET), URS or ESWL.
- **Obstructed:** any of
  - retropulsion to the kidney plus PCNL plus ureteroneocystostomy;
  - laparoscopic/robotic ureteroneocystostomy with stone retrieval;
  - URS with endoureterotomy (segment under 3 cm);
  - open ureteroneocystostomy with stone retrieval.

Complications of PCNL

▸ **Clavien-Dindo classification (PCNL complications)**

- **Rates:** none (grade 0) 77%, grade I 11%, grade II 7%, grade III 4.1%, grade IV 0.6%.
- **Grade I:** allows antiemetics, antipyretics, analgesics, diuretics, electrolytes and physiotherapy.
- **Grade II:** includes total parenteral nutrition and blood transfusion.
- **Grade IIIa / IIIb:** intervention without / under general anaesthesia.
- **Grade IVa / IVb:** single-organ dysfunction (including dialysis) / multi-organ dysfunction.

CH 98 · SURGICAL MANAGEMENT OF UPPER URINARY TRACT CALCULI

Evaluation of outcome: residual fragments

▸ **Clinically insignificant residual fragments (CIRF)**

- Fragments of 4 mm or less, non-obstructive and non-infectious, with sterile urine, in an asymptomatic patient.

▸ **Effectiveness coefficient (reporting outcomes)**

- Stone-free rate; success rate (stone-free rate plus CIRF); effectiveness coefficient.
- **Effectiveness coefficient** (for comparing modalities): % stone free divided by (100% + % retreatment + % auxiliary procedures), multiplied by 100.

CH 99 · PREVENTION OF URINARY STONE DISEASE

Treatment of hypercalciuria

▸ **Sodium cellulose phosphate**

- Non-absorbable resin that binds calcium and inhibits its absorption; no longer used.

▸ **Orthophosphate**

- Sodium or potassium salt of phosphorus; may have a role when other treatments fail.
- **Action:** inhibits vitamin D synthesis and directly alters renal tubular calcium reabsorption.
- **Problems:** parathyroid stimulation, soft-tissue calcification, marked rise in urinary phosphorus, increased urinary brushite saturation, GI distress, hypomagnesaemia and hyperoxaluria (check).
- May be used in mild to moderate resorptive hypercalciuria.

Ammonium acid urate stones

- Alkalinisation plus the common therapies: fluids, moderate calcium intake and a low-oxalate diet.
- Correct the cause: encourage healthy bowel habits in laxative abuse; rectify bowel disease or chronic diarrhoea if possible.

CH 125 · GENITAL AND LOWER URINARY TRACT TRAUMA

Posterior urethral injury

▸ **Female urethral injury**

- Needs immediate primary repair, or at least realignment over a catheter, to prevent later urethrovaginal fistula or urethral obliteration.
- Suture vaginal lacerations to prevent vaginal stenosis.
- Delayed reconstruction is very difficult, because the small urethra is embedded in scar and hard to mobilise.

CH 62 · PAEDIATRIC GENITOURINARY TRAUMA

Ureteral strictures

▸ **Davis intubated ureterotomy**

- Used for very long strictures; a modern modification adds a buccal mucosa graft.

OTHER TOPICS IN THIS BOOK

Lasers

▸ **CO2 laser**

- Infrared gas laser emitting a continuous wave.
- **Delivery:** rigid articulated arm rather than a fibre.
- **Tissue effects:** vaporisation and coagulation.
- **Other indications:** skin resurfacing, vocal cord lesions.

▸ **Erbium:YAG laser (versus holmium) for lithotripsy**

- Er:YAG is the second laser (with holmium) used for lithotripsy.
- **Er:YAG wavelength:** 2940 nm.

Feature	Holmium	Er:YAG
Mechanism	Photothermal vaporisation	Optical (photomechanical) energy
Shock wave	Weak	Strong
Vapour bubble	Pear-shaped, more heat loss	Torpedo-shaped
Soft-tissue penetration	Greater	Smaller, safer
Fragmentation	Slow, fine fragments	Fast, large fragments
Pulse duration	Longer	Shorter
Standard silica quartz fibres	Compatible	Cannot carry the energy

▸ **Problems with Er:YAG fibres**

- **Main barrier:** no fibres with adequate mechanical, chemical and thermal reliability.
- **Sapphire:** too brittle and too thick.
- **ZBLAN fluoride glass:** low melting point, inadequate durability.
- **Hybrid** (fluoride core, sapphire tip, circumferential cladding): under evaluation.

Fetuin in stone formation

▸ **Fetuin**

- Plasma protein secreted by the liver; binds calcium and phosphorus and so inhibits biomineralisation.
- When calcium and phosphorus reach saturation, its inhibition is overcome and it precipitates as mineraloprotein complexes that grow into a nidus for stones.
- **Factors overcoming inhibition:** fetuin gene polymorphism, saturation levels of calcium and phosphorus, altered microenvironment such as pH.

ADDITIONS FOUND ON RE-CHECK

Dual-modality lithotripters

- Combine pneumatic and ultrasonic lithotripsy in one handpiece: the pneumatic part breaks harder stones; the ultrasonic part makes smaller fragments and removes them.

- **LithoClast Ultra:** the front piece holds the ultrasonic lithotripter, with a central channel for the pneumatic probe; the handpiece has three cords (ultrasound power, air and suction).
- **CyberWand:** disposable probes with an inner and an outer sheath; vibration passes from the inner to the outer sheath, which then moves ballistically like a pneumatic lithotripter.

12 Bladder Cancer & Diversion

CH 129 · TUMOURS OF THE URINARY BLADDER

Etiology: genetic factors

▸ **GSTM1 (glutathione-S-transferase M1)**

- **Role:** conjugates reactive chemicals such as arylamines and nitrosamines.
- Null polymorphism increases risk 1.5 times.

▸ **Slow NAT-2 with null GSTM1**

- High levels of 3-aminobiphenyl and a higher risk of bladder cancer.

Urothelium, precursor lesions and molecular pathways

▸ **Proliferation markers (PCNA, MIB-1)**

- PCNA and MIB-1 are found in high-grade tumours and indicate poor prognosis.
- PUNLMP has the same proliferation rate as low-grade urothelial carcinoma.

Recurrence, dissemination and prognosis

▸ **Pagetoid spread**

- **Definition:** spread of cancer cells underneath a layer of normal-appearing surface urothelium, seen mainly in urothelial CIS.
- **Frequency:** 15% of bladders with CIS and 11% of high-grade papillary NMIBC.
- Can extend into the prostatic urethra and distal ureters.

Cystoscopy and optical imaging

▸ **Confocal laser endomicroscopy**

- **Principle:** a low-power laser illuminates tissue and the emitted fluorescent light is detected, using topical or intravenous fluorescein, to a depth of about 120 µm.
- **Strength:** distinguishes high-grade from low-grade tumours.
- **Limits:** restricted view, needs contrast, cannot detect muscle invasion.

▸ **Raman shift spectroscopy**

- Named after the Indian physicist C.V. Raman.
- **Principle:** measures the energy photons gain or lose from intramolecular bonds in tissue. Emitted wavelengths are studied as wave numbers; the Raman shift is the change in photon energy.
- Can assess invasion and distinguish low-grade from high-grade tumour, but is still in trials.

▸ **UV autofluorescence**

- Ultraviolet excitation of tissue; detects invasion and distinguishes low-grade from high-grade tumours.
- **Drawback:** toxicity.

▸ **Multiphoton microscopy**

- **Principle:** absorption of two infrared photons (fluorescence imaging) to a depth of about 0.5 mm.
- Distinguishes benign from malignant tissue and allows cytological grading, but does not detect muscle invasion.

▸ **Scanning fibre endoscopy**

- Small flexible endoscope with laser-based imaging giving a panoramic view of the urothelium; still in trials.

▸ **Molecular imaging (anti-CD47)**

- **Definition:** visualises and characterises biological processes at the molecular and cellular level.
- **In the bladder:** fluorescently labelled anti-CD47 antibody is instilled, the bladder is emptied, then optical imaging with confocal laser endomicroscopy is done. Anti-CD47 binds only tumour cells, so fluorescent tissue is biopsied.

Urinary molecular markers

▸ **Microsatellite analysis**

- PCR amplification detects tumour-associated loss of heterozygosity.
- **Accuracy:** sensitivity 72-97%, specificity 80-100%; described as the most promising marker.

▸ **DNA methylation (CpG islands)**

- Sensitivity 75%, but methylated CpG islands can occur in normal urothelial cells of older patients.

▸ **Lewis blood group antigen X**

- No commercially available test.
- Usually absent from adult urothelial cells except occasional umbrella cells.

▸ **CK20 and CYFRA 21-1**

- Cytoskeletal proteins detectable in urine by protein or mRNA assays.
- CYFRA 21-1 is not a useful marker in its current form.

▸ **Survivin**

- Anti-apoptotic protein highly expressed in urothelial cancer cells.
- May predict response to intravesical therapy, but is relatively poor at detecting advanced-stage or high-grade tumours.

▸ **Hyaluronic acid, telomerase and BLCA-4**

- **Hyaluronic acid:** role in cell proliferation; urinary amount correlates with stage.
- **Telomerase:** activity detected in 80% of urine samples from bladder cancer patients.
- **BLCA-4:** nuclear transcription factor marker with high specificity and sensitivity.

CH 130 · MANAGEMENT STRATEGIES FOR NON–MUSCLE-INVASIVE BLADDER CANCER

Laser and office-based management

▸ **Laser ablation of bladder tumours**

- **Best probe:** end-fire non-contact fibre with 5–15 degree divergence, giving penetration up to 5 mm.
- White tissue means protein denaturation; stop at that point.
- **Drawbacks:** higher cost and no tissue for biopsy.

Interferon and other immunotherapy

▸ **Intravesical IFN-α**

- The most extensively studied intravesical interferon.
- As a single agent, more expensive and less effective than BCG or intravesical chemotherapy.
- **Use:** salvage agent; not used perioperatively.
- **Toxicity:** cystitis in under 5%; flu-like symptoms in 20%.

▸ **Other investigational agents**

- IL-2 and a (mycobacterial) cell wall DNA extract (check).

Photodynamic therapy

- **Role:** an alternative for refractory NMIBC.
- **Mechanism:** a photosensitiser such as hexaminolevulinate (HAL) is instilled; red laser light with molecular oxygen causes free-radical cytotoxic injury.
- **Agents:** porfimer sodium can be given systemically instead of intravesical HAL. Emerging agents are hypericin and radachlorin.
- **Light:** red laser at 630 nm for 12–20 minutes, 2–3 days after the photosensitiser, once it has cleared from normal tissue.
- **Side effects (limit use):** bladder contracture or irritability in 50%; dermal photosensitivity in 19%.

CH 131 · MANAGEMENT OF MUSCLE-INVASIVE AND METASTATIC BLADDER CANCER

Pelvic lymph node dissection (PLND)

▸ Lymph node density

- Positive nodes as a percentage of nodes removed. Below 20% is associated with better survival, though this is controversial.

Metastatic bladder cancer

▸ Vinflunine and other drugs tried

- **Vinflunine:** antitubulin vinca alkaloid, approved in Europe as a second-line agent.
- **Taxanes:** paclitaxel, docetaxel, larotaxel.
- **Antifolates:** pemetrexed and piritrexim (synthetic).
- **VEGF inhibitors (alone or in combination):** bevacizumab, sunitinib, sorafenib.
- **EGFR-directed:** cetuximab with or without paclitaxel or GC, gefitinib, lapatinib, erlotinib.
- **Others:** trastuzumab (anti-HER2), ipilimumab (CTLA-4 inhibitor), cabozantinib (anti-angiogenic).

CH 133 · USE OF INTESTINAL SEGMENTS IN URINARY DIVERSION

Intestinal anastomosis: types

▸ Compression anastomosis (biofragmentable ring)

- The biofragmentable ring has proved safe, effective and time-efficient.

Ureterocolic anastomosis

▸ Strickler ureterocolic anastomosis

- Non-refluxing ureterocolic tunnel; the ureter enters from the lateral side.

▸ Pagano ureterocolic anastomosis

- Both ureters through a single incision into two submucosal tunnels.
- Lowest stricture rate among ureterocolonic anastomoses.

▸ Cordonnier and Nesbit ureterocolic anastomoses

- Refluxing; not preferred for ureterosigmoidostomy.

Ureteroileal anastomosis: techniques

▸ Split-nipple (Griffiths)

- The spatulated ureter is turned back on itself and the nipple is implanted into the bowel.

▸ **Le Duc anastomosis**

- The ureter is laid onto the inner bowel wall and becomes a submucosal tunnel when the mucosa re-epithelialises over it.

▸ **Hammock anastomosis**

- The conjoined ureters are sutured to the intestinal mucosa, with longitudinal seromuscular incisions cross-hatched by multiple incisions.

▸ **Ureteral dipping**

- The ureter is dipped into the bowel lumen and secured to the seromuscular layer without a mucosal anastomosis.

▸ **U-SB anastomosis**

- Antireflux mechanism: serosal compression of an extramural ureter.

Rectal-bladder urinary diversion

▸ **Folded rectosigmoid bladder**

- Ureters implanted through serosa-lined tunnels.

▸ **Augmented valved rectum**

- A proximal sigmoid intussusception confines urine to a smaller surface area.

▸ **Rectal hemi-Kock and T-Kock**

- The ileal patch is anastomosed to the rectum.

▸ **Mainz II (sigma-rectum pouch)**

- Simple and reproducible. Partial reconfiguration of the rectosigmoid junction creates a low-pressure reservoir with increased capacity.

Continent catheterising pouches

▸ **Kock pouch (continent ileal reservoir)**

- 15 cm of terminal ileum with an intussuscepted nipple valve.
- Preserves the ileocaecal valve.
- High complication rate; abandoned.

▸ **Double T-pouch**

- 70 cm of terminal ileum in a W-shaped configuration.
- **Continence:** an antiperistaltic 12–15 cm segment of distal ileum.
- **Antireflux:** the proximal 10 cm.
- Preserves the ileocaecal valve; fewer complications than the Kock pouch.

▸ **Mainz I pouch**

- W configuration from limbs of about 10–15 cm, 10–15 cm, 10–15 cm and 20–25 cm (caecum/ascending colon and ileum).
- **Nipple valve:** intussusception of ileum through the ileocaecal valve.
- High complication rate (31%).

▸ **Penn pouch**

- Ileocaecal pouch with 10–12 cm of ileum; the first diversion to employ the Mitrofanoff principle.
- The appendix is excised with a button of caecum and reversed on itself before tunnelled reimplantation.
- **Alternative:** bury the appendix in the taenia with the tip brought out as the stoma.
- Easy to construct; suits patients without a previous appendicectomy.

▸ **Gastric continent cutaneous pouch**

- A wedge-shaped segment of stomach from the greater curvature, based on the right gastroepiploic artery (also recorded as the left gastroepiploic artery).
- **Continence:** the contralateral distal ureter (after transureteroureterostomy) tunnelled into the reservoir, like an appendix.
- **Alternative:** a Mitrofanoff procedure, with a composite gastric reservoir and ileum segment and an appendix channel.
- **Possible advantages:** fewer infections and fewer metabolic complications; usable when the bowel is irradiated or short bowel syndrome is anticipated.
- May be especially useful in paediatric patients, but experience is limited at present.

CH 134 · CONTINENT URINARY DIVERSION

Orthotopic reservoirs

▸ **Le Bag**

- Ileocolic: ileum, caecum and colon, about 20 cm of colon and 20 cm of ileum, opened on the antimesenteric border and sutured to form the posterior wall.
- Urethra joined end-to-side to the caecum, ureters anastomosed, neobladder closed anteriorly.

CH 135 · MINIMALLY INVASIVE URINARY DIVERSION

Alternatives to GI cystoplasty

▸ **Seromuscular enterocystoplasty**

- Good early results, technically more difficult, long-term results unknown.

▸ **Urinary buttons**

- Alternative to a conduit or suprapubic catheter for drainage.
- **Design:** internal part like a Foley tip with a balloon; external flat button on the skin.
- A one-way valve maintains continence; no stoma bag needed.

ADDITIONS FOUND ON RE-CHECK

Local invasion of bladder cancer

- Genetic and epigenetic changes produce substances that invade the basal lamina, connective tissue and angiolymphatic system: collagenases, motility and growth factors, and cell adhesion molecules.
- **Proepithelin:** promotes establishment and progression; may be a novel biomarker.
- **Growth factors:** epidermal growth factors, TGF-alpha, heparin-binding growth factor, insulin-like growth factor.
- **Adhesion molecules in invasive cancer:** E-cadherin, integrins, CD44.

13 Kidney & UTUC

CH 136 · BENIGN RENAL TUMOURS

Mixed epithelial and stromal tumour (MEST)

▸ **Pathogenesis (both)**

- Sex hormones play a role, suggested by female predilection, a history of hormonal ablation therapy in affected males, and frequent ER/PR expression.

▸ **MEST**

- Peaks in the 5th decade, usually in women with a history of oestrogen replacement therapy.
- Appears as a Bosniak III/IV cyst, indistinguishable from cystic RCC; consider it pre-operatively in a woman on HRT.
- **Treatment:** partial nephrectomy if feasible, otherwise radical nephrectomy.

Metanephric adenoma and papillary adenoma

▸ **Metanephric adenoma**

- Radiologically indistinguishable from RCC; shares some features with Wilms tumour.
- About 10% have polycythaemia.
- Peripheral or central calcification; enhances less than the surrounding cortex and medulla.
- Usually confirmed after excision: WT1 and S100 positive, and BRAF V600E mutation testing.

▸ **Papillary adenoma**

- May be premalignant for papillary RCC; many believe all solid epithelial renal masses are potentially malignant and need treatment.
- Incidence rises with age, male sex and acquired renal cystic disease.

CH 137 · MALIGNANT RENAL TUMOURS

Pathological subtypes

▸ **Fuhrman nuclear grade**

Grade	Nuclear size	Features
I	About 10 µm	Round, uniform; nucleoli absent or inconspicuous
II	About 15 µm	Irregular; small nucleoli visible at 400x
III	About 20 µm	Irregular; prominent nucleoli
IV	20 µm or more	Bizarre, multilobed; prominent nucleoli, heavy chromatin clumps

▸ **Post-neuroblastoma RCC**

- Post-neuroblastoma RCC is a separate pathological subtype.

Diagnosis

▸ **Renal arteriography**

- Now a limited role.
- **Indications:** renal mass with renal artery disease; equivocal cases where neovascularity may help; showing vascularity of IVC tumour thrombus (seen in 35–40%).

▸ **Inferior venacavography**

- Can assess IVC thrombus but is rarely done.

Staging

▸ **Robson staging**

Stage	Extent
I	Within the renal capsule
II	Into perinephric fat, confined within Gerota's fascia
III	Regional lymph nodes and/or renal vein or IVC
IV	Adjacent organs or distant metastasis

Management of localised RCC

▸ **Zonal NePhRO score**

- Four components, each scored 1, 2 or 3 by zone 1, 2 or 3.

Component	1 point	2 points	3 points
Nearness	Touches cortex	Touches medulla	Touches collecting system or crosses into renal sinus
Physical zone	Lower pole, below collecting system	Lateral to, not touching, pelvicalyceal system	Upper pole, or touching pelvicalyceal system
Radius	Under 2.5 cm	2.5–4 cm	More than 4 cm
Organisation	More than 50% exophytic	50–75% endophytic	More than 75% endophytic

- **Total:** 4–6 low, 7–9 intermediate, 10–12 high risk.

CH 138 · UROTHELIAL TUMOURS OF THE UPPER URINARY TRACT AND URETER

Diagnosis

▸ **Brush biopsy**

- Through a retrograde catheter or ureteroscope; sensitivity and specificity about 90%.
- **Complications:** massive haemorrhage, perforation, extravasation.

Prognostic factors

▸ **Broders grade**

Grade	Description
I	Papilloma
II	Well differentiated
III	Moderately differentiated
IV	Anaplastic or poorly differentiated

CH 141 · CONTEMPORARY OPEN SURGERY OF THE KIDNEY

Laparoscopic renal surgery: approaches and ports

▸ **NOTES (natural orifice transluminal endoscopic surgery)**

- Limited human experience; hybrid NOTES with laparoscopic assistance, using the vagina for access and extraction at nephrectomy, has been described.
- Transvesical nephrectomy recently done in animals.

CH 144 · TREATMENT OF ADVANCED RENAL CELL CARCINOMA

Immunotherapy and transplant

▸ **Allogeneic haematopoietic stem cell transplant**

- Replaces the recipient's immune and haematopoietic systems with those of a healthy HLA-compatible donor; acts by a graft-versus-tumour (allogeneic antitumour) effect.

Targeted therapy

▸ **Other VEGF-pathway agents: nintedanib and dovitinib**

- Nintedanib, dovitinib.

OTHER TOPICS IN THIS BOOK

UTUC: additional points

▸ **PCN-tract brachytherapy**

- **PCN-tract brachytherapy** after percutaneous organ-sparing therapy: uses iridium wire; risk of cutaneous fistula, which then needs RNU.

14 Adrenal & Retroperitoneum

CH 146 · PATHOPHYSIOLOGY, EVALUATION AND MEDICAL MANAGEMENT OF ADRENAL DISORDERS

Primary aldosteronism (Conn syndrome)

▸ **Mineralocorticoid escape**

- Escape from the sodium-retaining effects of excess aldosterone.
- Mediated by pressure natriuresis and atrial natriuretic peptide (ANP).
- Limits volume expansion to 1.5 kg or less.

▸ **Fludrocortisone suppression test**

- **Fludrocortisone suppression test:** once the gold standard, now out of favour.
  - Fludrocortisone 0.1 mg 6-hourly with NaCl 2 g 8-hourly for 4 days.
  - Plasma aldosterone concentration (PAC) then measured upright; PAC above 6 ng/dL is diagnostic.

▸ **NP-59 scintigraphy**

- **Tracer:** I-131 6β-iodomethyl-norcholesterol, a cholesterol-based cortisol analogue that labels adrenocortical cells.
- **Thyroid protection:** saturated potassium iodide (Lugol's solution) before injection, against uptake of free I-131.
- **ACTH suppression:** dexamethasone 1 mg 6-hourly for 7 days.
- **Timing:** imaging begins on day 4 after injection and may continue through day 10.
- **Interpretation:** unilateral uptake within 5 days suggests adenoma; bilateral uptake suggests idiopathic hyperplasia.
- Detection is proportional to adenoma size and function; SPECT-CT improves sensitivity for smaller adenomas.

▸ **Posture stimulation test**

- Aldosterone measured after overnight recumbency and after 4 hours upright.
- **Normal:** upright position raises angiotensin and aldosterone 2–4 times above the recumbent level.
- **Idiopathic hyperplasia:** aldosterone rises by at least 33% over baseline on standing.
- **Aldosterone-producing adenoma or FH type I:** no change with posture.
- **Limits:** no information on location; 30–50% of adenomas show a rise in PAC, and 20% of idiopathic hyperplasias show no rise.

▸ **18-hydroxycorticosterone (18-OHB)**

- Immediate precursor of aldosterone.
- Above 100 ng/dL suggests adenoma; below 100 ng/dL suggests hyperplasia.
- Accuracy below 80%; no information on side (right versus left).

Cushing syndrome

▸ **McCune–Albright syndrome and Carney complex**

- **McCune–Albright syndrome:** AIMAH, fibrous dysplasia, skin problems, other endocrine abnormalities.
- **Carney complex:** autosomal dominant; PPNAD, spotty skin and mucous membrane lesions, multiple tumours including Sertoli cell tumours.

▸ **Exogenous Cushing syndrome**

- Taper the steroid gradually when possible.

- **Steroid withdrawal syndrome:** the patient cannot tolerate dose reduction even though HPA axis tests appear normal.

▸ **Trans-sphenoidal surgery**

- **Relapse:** about 25%, so long-term follow-up is needed.
- **Macroadenoma:** 15% or fewer cured.
- Severe Addisonian state is common; glucocorticoid replacement needed for the year after surgery.
- **Hypopituitarism:** under 5% to 50%, a wide range.
- **Re-operation:** challenging, variable success, higher risk of hypopituitarism.
- **Salvage pituitary radiation:** ACTH falls slowly; hypopituitarism may follow.

Adrenal incidentaloma

▸ **11C-metomidate**

- **11C-metomidate:** inhibits 11β-hydroxylase and aldosterone synthase; preferentially taken up by adrenocortical cells; cannot separate benign lesions from adrenocortical carcinoma.

Phaeochromocytoma

▸ **Clonidine suppression test**

- Clonidine (alpha-2 agonist) suppresses sympathetic catecholamine (especially norepinephrine) production but not that from a phaeochromocytoma.
- Normetanephrine is compared before and after clonidine.
- **Main problem:** significant hypotension.

▸ **Chromogranin A**

- A granin, found in secretory vesicles of neuroendocrine and nervous tissue.
- **Sensitivity:** about 85%.
- **Use:** when free plasma metanephrines are mildly to moderately raised (less than 4-fold).
- Renally cleared, so specificity falls when GFR is below 80 mL/min.

▸ **CVD chemotherapy**

- **CVD chemotherapy:** short-lived response (under 2 years); radiographic shrinkage 50%, biochemical response 75%. Given mainly after MIBG failure or with no MIBG uptake initially.

CH 140 · RETROPERITONEAL TUMOURS

Retroperitoneal sarcoma

▸ **MAID regimen**

- **MAID:** mesna, doxorubicin (Adriamycin), ifosfamide, dacarbazine.

15 Renal, Nephrology & Transplant

CH 88 · SURGICAL, RADIOLOGIC AND ENDOSCOPIC ANATOMY OF THE KIDNEY AND URETER

Lumbodorsal fascia

- Deep investing membrane over most of the posterior thorax and abdomen; only a thin fibrous lamina in the thorax.
- **Insertions:** anterior and middle layers onto the vertebral transverse processes; posterior layer onto the spinous processes.
- **Laterally:** bridges the aponeuroses of internal oblique and transversus abdominis, the costal angles and the iliac crest to the medial vertebral column and sacrum.
- **Relations:**
  - Superficial to posterior layer: serratus posterior inferior and latissimus dorsi.
  - Between posterior and middle layers: erector spinae.
  - Between middle and anterior layers: quadratus lumborum.
  - Anterior to anterior layer: psoas major.

CH 91 · RENAL TRANSPLANTATION

Post-transplant infections

▸ **Vaccination**

- Any live vaccine at least 4 weeks before transplant.
- **Before transplant:** HPV, hepatitis B, pneumococcus.
- **Barred after (live):** polio, MMR.
- **Safe after:** hepatitis A and B, diphtheria, tetanus.

▸ **Timeline**

- **First month:**
  - Bacterial: UTI, wound infection, aspiration pneumonia, intra-abdominal infections, catheter and drain-site infections. Multidrug-resistant bacteria are an important cause of mortality and morbidity.
  - Viral: mainly HSV; others uncommon.
  - Fungal: Candida, including azole-resistant Candida.
- **1 to 6 months (opportunistic, from immunosuppression):**
  - Viral: CMV, HHV-6, HSV, VZV, EBV, HBV, HCV, BK virus.
  - Fungal: Aspergillus, Cryptococcus, mucormycosis.
  - Bacterial: Nocardia, TB, Legionella.
  - Parasitic: Pneumocystis jirovecii, Toxoplasma, Leishmaniasis.
- **After 6 months:**
  - Good graft function, no chronic viral infection, low-dose immunosuppression: mainly community-acquired infections.
  - Chronic viral infection: progressive liver disease (HBV/HCV), BK nephropathy or PTLD (EBV).
  - Multiple rejection episodes: opportunistic bacterial, viral and fungal infections.

▸ **Prophylaxis**

Target	Prophylaxis
UTI	TMP-SMX for 3–6 months, or nitrofurantoin
Pneumocystis pneumonia	TMP-SMX, or inhaled pentamidine

Oral candidiasis	Nystatin suspension for 1–3 months, or clotrimazole lozenges (check)
Vaginal candidiasis	Intravaginal nystatin or clotrimazole (check)
HSV/CMV	Acyclovir or valacyclovir for 3 months
Peptic ulcer	H2-receptor blocker, plus a second agent if symptomatic

▸ **Cytomegalovirus (CMV)**

- **Timing:** mainly 1–6 months after transplant.
- **Patterns:** primary infection (recipient negative, donor positive); reactivation of latent virus (recipient positive, donor positive or negative); superinfection with a new virus in a seropositive recipient (both positive).
- **Effects:** any organ (for example hepatitis, retinitis); lowers platelet and total leucocyte counts; raises the risk of opportunistic infection and PTLD.
- **Prevention:** acyclovir or valganciclovir.
- **Diagnosis:** CMV PCR assay.
- **Treatment:** IV ganciclovir; ganciclovir-resistant disease with cidofovir or foscarnet.

OTHER TOPICS IN THIS BOOK

Acid-base regulation

- **Three systems:** kidney, lungs and buffers.
- **Kidney:** distal H+ secretion alone would drop urine pH below 2.5, toxic to tubular cells, so the acid is buffered with ammonium in collecting duct cells.
- **Proximal tubule:** luminal H+ (secreted in exchange for Na+) combines with filtered HCO3 to form H2CO3; carbonic anhydrase converts it to CO2 and water, which enter the cell; carbonic anhydrase regenerates H+ and HCO3, and the HCO3 passes to blood.
- **Lungs:** exhale acidic CO2; a rise in blood CO2 drives hyperventilation, removing CO2.
- **Buffers:** the body's weak acids and bases, guarding against sudden pH shifts. Most important in blood: carbonic acid and bicarbonate.

Normal values and disorders

- **Normal arterial values:** pH 7.35-7.45, pO2 80-100 mmHg, pCO2 35-45 mmHg, HCO3 22-28 mEq/L.
- **Disorders:** metabolic acidosis, renal tubular acidosis, metabolic alkalosis, respiratory acidosis, respiratory alkalosis.

Disorder	Primary change	Compensation
Metabolic acidosis	HCO3 falls	pCO2 falls
Metabolic alkalosis	HCO3 rises	pCO2 rises
Respiratory acidosis	pCO2 rises	HCO3 rises
Respiratory alkalosis	pCO2 falls	HCO3 falls

Metabolic alkalosis

- **Definition:** HCO3 above 26 mEq/L.
- **Causes:** excessive vomiting, certain diuretics, severe dehydration, excessive alkali drugs, heavy antacid ingestion, endocrine disorders.
- **Treatment:** mainly volume and electrolyte replacement and treating the cause; in severe cases occasionally dilute acid as intravenous ammonium chloride.

17 Prostate Cancer

CH 153 · EPIDEMIOLOGY, AETIOLOGY AND PREVENTION OF PROSTATE CANCER

Chemoprevention trials

▸ **Toremifene citrate**

- Oestrogen receptor modulator, FDA-approved for breast cancer.
- Its RCT failed to show a reduction in prostate cancer.

CH 154 · PROSTATE CANCER BIOMARKERS

Biomarker development phases (EDRN)

- **Early Detection Research Network (EDRN):** set up to identify, develop and validate promising biomarkers and technologies for early cancer detection.

Phase	Content
1	Biomarker discovery in biomarker development laboratories
2	Sensitivity and specificity; ability to tell cases from controls
3	Ability to detect preclinical disease; lead time before clinical presentation
4	Indications for applying the biomarker clearly defined
5	Cancer control studies: does it reduce the burden of disease? Reduced mortality is the successful outcome

Assessment of biomarker performance

▸ **Sensitivity and specificity**

- **Sensitivity:** proportion of people with the disease who test positive. TP divided by (TP plus FN), times 100.
- **Specificity:** proportion of people without the disease who test negative. TN divided by (TN plus FP), times 100.

▸ **Accuracy and the ROC curve**

- **Accuracy:** (TP plus TN) divided by the total population.
- **Area under the ROC curve (AUC):** measures accuracy. 1 is an ideal test; 0.5 is a poorly performing biomarker.
- **Axes:** sensitivity on the vertical axis, 1 minus specificity on the horizontal axis.

▸ **Predictive values**

- **PPV:** probability that a positive test indicates disease. TP divided by (TP plus FP).
- **NPV:** probability that a negative test indicates absence of disease. TN divided by (TN plus FN).
- NPV and PPV are more relevant to clinical practice than sensitivity and specificity.

▸ **Likelihood ratio**

- How much a new biomarker raises or lowers the probability of disease before biopsy.

Regulatory status

- **FDA-approved:** PSA, free-to-total PSA ratio, PHI and PCA3.
- **CLIA certification (Clinical Laboratory Improvement Amendments):** allows a test to be offered commercially, but does not guarantee rigorous clinical evaluation.

Human kallikrein 2 (hK2) and other kallikreins

- 80% amino acid homology with PSA; similar specificity for prostatic tissue; regulated by androgens.
- **Concentration:** less than 2% of total PSA in prostatic tissue, seminal fluid and serum.
- **Serum forms:** bound and free, mostly free.
- **Expression:** varies independently of PSA; less intense than PSA in benign tissue and more intense in cancer.
- Correlates with more aggressive pathological features.
- **Other kallikreins:** KLK4, KLK14 and KLK15, found mainly in tissue rather than serum.

Endoglin

- CD105; transmembrane glycoprotein and cell-surface coreceptor for TGF-beta 1 and TGF-beta 3, on vascular endothelial cells; key role in angiogenesis.
- **Pre-operative plasma level:** associated with node-positive disease; predicts biochemical recurrence.
- **Urinary level:** correlates with tumour volume.
- Used alone or with other markers; clinical utility needs further study.

Circulating tumour cells (CTCs)

- **Definition:** CD45-negative cells positive for EpCAM and/or cytokeratin.
- No single gold-standard identification method.
- **CellSearch:** the only FDA-approved method. Antibodies against EpCAM, CD45 (negative) and cytokeratins 8, 18 and 19 (positive).
- **Prognosis:** 5 or more CTCs per 7.5 mL of blood is poor prognosis in prostate, colorectal and breast cancer.
- **CTC-chip:** developed at Harvard Medical School; highest sensitivity; detects a single EpCAM-positive cell among 1 billion blood cells.

Autoimmune responses

- The body's autoimmune response to cancer (autoimmune signatures) can be a target for biomarker development.

Urine-based biomarkers

▸ Metabolomics (sarcosine)

- Exploits differences in metabolic processes between cancer cells and benign cells.
- **Sarcosine:** the most prominent metabolite; a metabolite of glycine found in the urine of patients with prostate cancer.
- **Prostarix:** commercial panel run on a post-DRE sample; not FDA approved.

Other serum and tissue biomarkers

▸ Annexin A3

- Calcium- and phospholipid-binding protein; level inversely related to presence of prostate cancer; used alone or with other markers.

▸ MicroRNA (miR-141)

- Single-stranded RNAs that regulate mRNA, with diagnostic and prognostic roles. miR-141 is a marker for prostate cancer.

▸ AMACR (alpha-methylacyl coenzyme A racemase)

- **Gene:** chromosome 5; upregulated in prostate cancer.
- **Function:** beta-oxidation of fatty acids.

- **In biopsy tissue:** 97% sensitivity and 100% specificity.

CH 155 · PROSTATE BIOPSY: TECHNIQUES AND IMAGING

Transperineal and transurethral biopsy

▸ **Transurethral biopsy**

- Not done, because TRUS biopsy samples the transition zone adequately.
- Solitary transition-zone cancers without peripheral-zone tumour are rare (under 5%).

Advanced and investigational techniques

▸ **Elastography**

- Measures tissue stiffness; reduced elasticity suggests malignancy. May prove better than colour Doppler at finding malignant areas.

▸ **Prostate HistoScanning**

- Evolving technology that differentiates, characterises and displays prostate tissue by analysing back-scattered ultrasound.

▸ **Computerised biopsy templates**

- Turn 2D ultrasound data into a 3D model for targeted imaging (TargetScan, Artemis).

CH 157 · DIAGNOSIS AND STAGING OF PROSTATE CANCER

Whitmore–Jewett staging system

Whitmore–Jewett stage	Description
A	Found incidentally; not clinically detectable
B	Found on physical examination; not spread outside the capsule
C	Extended through the capsule
D	Spread to other organs

CH 160 · OPEN RADICAL PROSTATECTOMY

Modifications of classical RP

▸ **Bladder neck sparing RP**

- **Aim:** hasten return of urinary control.
- **Continence:** clinically significant advantage at 6 months, which largely disappears by 1 year.
- **Margins:** little difference in positive margin rates.
- Bladder neck contracture rates may be high, so not done routinely.
- **Heidelberg technique:** preserves the circular fibres of the bladder neck, with a urethro-urethral anastomosis.

▸ **Seminal vesicle sparing RP**

- Most men with low-risk disease can undergo it.
- Non-oncological risks and benefits are uncertain and compromised cancer control is a concern; randomised trials are needed, so not done routinely.

CH 162 · RADIATION THERAPY FOR PROSTATE CANCER

Heavy particle beams

▸ **Heavy particle beam therapy**

- Includes neutron and proton beams. Limited availability and expensive; perceived higher complication rates without much gain in tumour control.

▸ **Bragg peak**

- The peak on the Bragg curve, which plots energy loss of ionising radiation as it travels through matter. The proton beam is widened (spread out) to treat a larger tumour volume.

CH 165 · MANAGEMENT OF RECURRENT AND NEWLY METASTATIC PROSTATE CANCER

LHRH antagonists

▸ **Abarelix**

- First-generation; causes a severe allergic reaction not seen with third- and fourth-generation antagonists.
- Observe for 30 minutes after injection.

Oestrogens

▸ **Diethylstilbestrol (DES)**

- Oestrogen that suppresses testosterone; about a thousand-fold more potent than testosterone at suppressing LH and FSH.
- No bone loss, but severe thromboembolism even at low doses and cardiovascular toxicity, so not standard first-line treatment.

Trials in hormone-sensitive disease

▸ **STAMPEDE trial**

- **Name:** Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy.
- **Design:** multi-arm, multi-stage (MAMS) randomised controlled trial in a tertiary-care setting.
- **Participants:** men starting long-term hormone therapy.
- **Hypothesis:** research interventions would improve survival over standard of care alone.
- **Primary outcomes:** overall survival (definitive) and failure-free survival (intermediate).
- **Secondary outcomes:** toxicity, quality of life, cost-effectiveness and symptomatic skeletal events.

Arm	Treatment	Result
A (control)	Standard of care: ADT, with or without RT, with or without docetaxel	Not applicable
B	Plus zoledronic acid	No increase in overall survival
C	Plus docetaxel	Overall survival up by 10 months; more side effects
D	Plus celecoxib	No change in overall survival
E	Plus zoledronic acid and docetaxel	Increased overall survival
F	Plus zoledronic acid and celecoxib	No change in overall survival
G	Plus abiraterone acetate and prednisolone	Increased overall survival, progression-free survival and time to radiographic progression
J	Plus abiraterone and enzalutamide	In follow-up
K	Plus metformin	Recruiting

L	Transdermal oestradiol, with or without RT, with or without docetaxel	Recruiting

CH 166 · MANAGEMENT OF CASTRATION-RESISTANT PROSTATE CANCER

Androgen synthesis inhibitors

▸ **Aminoglutethimide**

- **Action:** blocks side-chain cleavage enzyme, so cholesterol is not converted to pregnenolone.
- Causes medical adrenalectomy (blocks aldosterone and cortisol too), so fludrocortisone and cortisone replacement are needed.
- **Dose:** 1000 mg/day, the same as abiraterone.
- **Side effects:** anorexia, skin rash, lethargy, vertigo, nystagmus and hypothyroidism.

Targeted treatments

▸ **Custirsen**

- IV antisense oligonucleotide inhibiting clusterin, an anti-apoptotic protein; may reverse taxane resistance.
- **Trials:** SYNERGY and AFFINITY.

OTHER TOPICS IN THIS BOOK

Biases in PSA screening

▸ **Lead-time bias**

- **Compensation:** include the lead time when comparing screened and unscreened cohorts, and measure disease-specific mortality rather than survival time.

▸ **Length-time bias**

- Aggressive tumours have a short potential screening period (between possible screen detection and symptoms); slow-growing tumours have a longer one.
- Aggressive tumours therefore tend to present clinically between rounds, so screen-detected tumours appear to survive better; this reflects biology but is wrongly credited to screening, and is stronger with regular-interval screening.
- Overdiagnosis is its extreme form.